Data Model Design (DMD)

Project Metadata

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- **Domain:** Sales & Commercial Analytics
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1. Executive Summary & Purpose

The purpose of this document is to define and detail the logical, physical, and analytical architecture of the data model supporting the AdventureWorks Enterprise Sales Analytics Platform.

To ensure optimal query performance, report responsiveness, and intuitive self-service analytics, this platform adopts a **Dimensional Modeling** approach using a highly optimized **Star Schema** design. This model integrates transaction-level details from both B2C (Internet Sales) and B2B (Reseller Sales) channels, aligning them with internal corporate Sales Quotas to form a cohesive, enterprise-wide semantic layer in Power BI.

2. Modeling Methodology & Design Principles

The data architecture is engineered in accordance with the industry-standard **Ralph Kimball Dimensional Modeling Methodology**.

2.1 Galaxy Schema

- **Denormalization:** Dimensions (such as Product and Geography) are flattened (denormalized) into single-level tables directly connected to fact tables. This eliminates snowflake-like multi-hop joins (e.g., Product $\rightarrow$ Subcategory $\rightarrow$ Category), drastically reducing query processing times in DAX (Data Analysis Expressions) and simplifying the user experience for ad-hoc drag-and-drop analysis.
- **Direct Joins:** All dimension tables join directly to the relevant fact tables via single-column primary-to-foreign key relationships.

2.2 Conformed Dimensions

To enable cross-channel and cross-process analytical capabilities, we employ **Conformed Dimensions** (e.g., DimDate, DimProduct, DimSalesTerritory, DimEmployee). These shared dimensions guarantee that filtering by attributes like "Calendar Year" or "Product Category" yields perfectly aligned and comparable metrics across B2C Sales, B2B Sales, and Sales Quota plans.

2.3 Semantic Design Goals

1. **Performance Maximization:** Minimize column counts, eliminate redundant data types, and utilize integer-based keys for high-speed VertiPaq in-memory storage.
2. **Intuitive Self-Service:** Provide business users with clean, readable attribute names (e.g., Product Name instead of EnglishProductName) grouped into logical folders.
3. **Reusable DAX Logic:** Ensure that all primary metrics are defined globally, preventing regional teams from creating inconsistent local formulations.

3. Business Process Coverage

The core sales domain of AdventureWorks is divided into three distinct business processes, each mapped to its primary transactional system and fact table:

Business Process	Source Transaction System	Primary Fact Table	Granularity (Grain)
Internet Sales (B2C)	E-Commerce Platform Database	FactInternetSales	One row per product SKU per sales order line.
Reseller Sales (B2B)	ERP Sales Ledger	FactResellerSales	One row per product SKU per reseller purchase order line.
Sales Target Management	FP&A Planning Sheets	FactSalesQuota	One row per sales representative per fiscal quarter.

4. Fact Tables Detail

4.1 FactInternetSales

- **Business Process:** Direct consumer transactions completed via the online retail storefront.
- **Granularity:** Line-item level (Individual product items within a unique customer order).

A. Foreign Key Mappings

Key Name	Connected Dimension Table	Description
ProductKey	DimProduct	Maps the specific SKU purchased.
CustomerKey	DimCustomer	Identifies the B2C consumer demographic profile.
PromotionKey	DimPromotion	Maps any active marketing campaign or discount applied.
CurrencyKey	DimCurrency	Captures original transaction currency for multi-currency conversion.
SalesTerritoryKey	DimSalesTerritory	Assigns the operational retail territory responsible for the sale.
OrderDateKey	DimDate	**Active Role:** Represents the day the order was placed.
ShipDateKey	DimDate	**Inactive Role:** Represents the day physical goods were shipped.
DueDateKey	DimDate	**Inactive Role:** Represents the scheduled delivery date.

B. Fact Measures (Numerical Fields)

- SalesAmount (Gross revenue generated from the line item)
- OrderQuantity (Units purchased)
- UnitPrice (Individual catalog item price)
- ExtendedAmount (Price before discounts: $\text{OrderQuantity} \times \text{UnitPrice}$)
- TotalProductCost (Standard cost of production/acquisition)
- TaxAmt (Tax applied to the line transaction)

- Freight (Allocated logistics/shipping cost)

4.2 FactResellerSales

- **Business Process:** Wholesale purchase transactions executed by authorized distributors and B2B retailers.
- **Granularity:** Line-item level (Individual product items within a corporate reseller purchase order).

A. Foreign Key Mappings

Key Name	Connected Dimension Table	Description
ProductKey	DimProduct	Maps the wholesale SKU distributed.
ResellerKey	DimReseller	Identifies the purchasing corporate entity (Distributor/Store).
EmployeeKey	DimEmployee	Identifies the Sales Representative handling the corporate account.
PromotionKey	DimPromotion	Track wholesale trade discounts or volume campaigns.
CurrencyKey	DimCurrency	Identifies the reporting/transaction currency.
SalesTerritoryKey	DimSalesTerritory	Links the commercial geographic division.
OrderDateKey	DimDate	**Active Role:** Represents the purchase order creation date.
ShipDateKey	DimDate	**Inactive Role:** Represents the bulk distribution shipment date.
DueDateKey	DimDate	**Inactive Role:** Represents

		the buyer's requested delivery deadline.

B. Fact Measures (Numerical Fields)

- SalesAmount (Wholesale revenue generated after standard discounts)
- OrderQuantity (Bulk units purchased)
- UnitPrice (Negotiated B2B item price)
- ExtendedAmount (Gross line item value before taxes and discounts)
- TotalProductCost (Wholesale product manufacturing and supply cost)
- TaxAmt (Commercial taxes applied)
- Freight (Bulk shipping fees)

4.3 FactSalesQuota

- **Business Process:** Management and allocation of performance targets to internal sales employees.
- **Granularity:** Period-based planning (One target record per individual sales representative per fiscal quarter).

A. Foreign Key Mappings

Key Name	Connected Dimension Table	Description
EmployeeKey	DimEmployee	Links the target to a specific active Sales Representative.
DateKey	DimDate	Binds the quota to a specific calendar date (typically first day of fiscal quarter).

B. Fact Measures (Numerical Fields)

- SalesAmountQuota (Established revenue milestone/target for the designated period: Q_{target})

5. Dimension Tables Detail

Dimension tables provide descriptive context (Group-by attributes) for the metrics stored in the fact tables.

5.1 DimDate

- **Purpose:** Standardizes time-intelligence reporting, fiscal calendar alignment, and historical trend analysis across all corporate platforms.
- **Primary Key:** DateKey (Integer representation: YYYYMMDD)
- **Key Attributes:**
  - FullDateAlternateKey (True Date Format)
  - DayNumberOfWeek, EnglishDayNameOfWeek (Weekly analysis)
  - MonthNumberOfYear, EnglishMonthName (Monthly grouping)
  - CalendarQuarter, CalendarYear (Standard corporate cycles)
  - FiscalQuarter, FiscalYear (AdventureWorks-specific planning periods)

5.2 DimProduct (Consolidated Star Dimension)

- **Purpose:** Provides descriptive details for products sold. To avoid snowflaking, this dimension has been de-normalized to include metadata from subcategory and category levels.
- **Primary Key:** ProductKey
- **Conformed Hierarchical Attributes:**
  - EnglishProductName (Individual SKU descriptive name)
  - Color (Product surface styling)
  - Size (Physical dimensions/frame sizes)
  - ModelName (Overarching design model)
  - ProductSubcategoryName (Consolidated from downstream table)
  - ProductCategoryName (Consolidated from downstream table)
  - Status (Active, Obsolete, Current)

Hierarchy Path:
[Product Category] ──> [Product Subcategory] ──> [Product Model] ──> [Product Name / SKU]

5.3 DimCustomer

- **Purpose:** Stores demographic profiles of direct B2C internet shoppers.
- **Primary Key:** CustomerKey
- **Key Attributes:**
  - FirstName, LastName (Customer Identity)
  - Gender (Demographic targeting)
  - YearlyIncome (Affluence segmentation)
  - MaritalStatus, Education, Occupation (Behavioral profiling)
  - GeographyKey (Spatial relationship to regional tables)

5.4 DimReseller

- **Purpose:** Stores structural and financial attributes of wholesale distributors and corporate partner accounts.

- **Primary Key:** ResellerKey
- **Key Attributes:**
  - ResellerName (Company legal name)
  - BusinessType (e.g., Value Added Reseller, Warehouse, Specialty Bike Shop)
  - ProductLine (Focus catalog segment)
  - AnnualRevenue, NumberEmployees (Corporate scale indicators)

5.5 DimEmployee

- **Purpose:** Stores structural profiles of AdventureWorks staff, with an emphasis on sales representatives and organizational reporting hierarchies.
- **Primary Key:** EmployeeKey
- **Key Attributes:**
  - FirstName, LastName (Representative Identity)
  - Title (Corporate rank)
  - BaseRate, SalesPersonFlag (Operational filters)
  - ParentEmployeeKey (Self-referencing foreign key for organizational hierarchy modeling)

5.6 DimPromotion

- **Purpose:** Evaluates marketing campaign effectiveness, volume discounts, and seasonal sales strategies.
- **Primary Key:** PromotionKey
- **Key Attributes:** EnglishPromotionName, DiscountPct, EnglishPromotionType, EnglishPromotionCategory.

5.7 DimCurrency

- **Purpose:** Maps multi-currency cash flows to standard transactional baselines.
- **Primary Key:** CurrencyKey
- **Key Attributes:** CurrencyAlternateKey (ISO standard code, e.g., USD, EUR), CurrencyName.

5.8 DimSalesTerritory

- **Purpose:** Defines corporate sales boundaries, facilitating territory reporting and commission alignment.
- **Primary Key:** SalesTerritoryKey
- **Key Attributes:** SalesTerritoryRegion, SalesTerritoryCountry, SalesTerritoryGroup.

5.9 DimGeography

- **Purpose:** Normalizes spatial identifiers. Directly joined to Customer and Reseller tables to track regional distributions.
- **Primary Key:** GeographyKey

- **Key Attributes:** City, StateProvinceName, EnglishCountryRegionName, PostalCode.

6. Date Roles (Role-Playing Dimensions)

Both FactInternetSales and FactResellerSales contain multiple date fields (OrderDateKey, ShipDateKey, DueDateKey) referencing DimDate. To maintain clean star schema relationships without duplication, **Date Role-Playing** is implemented:

```
        ┌─────────────┐
        │  DimDate    │
        └──────┬──────┘
               │
   ┌───────────┼───────────┐
 Active     Inactive     Inactive
   ▼           ▼           ▼
[OrderDate] [ShipDate]  [DueDate]
```

1. **Active Relationship:** * Linked via OrderDateKey to DimDate.DateKey. All standard date filters automatically apply to the transactional order date.
2. **Inactive Relationships:** * Linked via ShipDateKey and DueDateKey to DimDate.DateKey.
   - **Activation Method:** These relationships remain dormant and are activated programmatically using DAX USERELATIONSHIP() when computing fulfillment rates or shipping lag metrics.

Example DAX Activation for Shipping Amount:

```
Shipped Revenue =
CALCULATE(
  [Total Revenue],
  USERELATIONSHIP(FactInternetSales[ShipDateKey], DimDate[DateKey])
)
```

7. Conformed Dimensions Matrix

To guide BI developers in identifying safe metrics and cross-process evaluation paths, the following matrix maps dimensional validity across the transactional and planning fact tables:

Dimension Table	**FactInternetSales (B2C)**	**FactResellerSales (B2B)**	**FactSalesQuota (Targets)**
DimDate	✅ (Conformed)	✅ (Conformed)	✅ (Conformed)

DimProduct	✅ (Conformed)	✅ (Conformed)	❌ (Not Applicable)
DimSalesTerritory	✅ (Conformed)	✅ (Conformed)	❌ (Indirect via Employee)
DimEmployee	❌ (Not Applicable)	✅ (Direct Joint)	✅ (Direct Joint)
DimCustomer	✅ (Direct Joint)	❌ (Not Applicable)	❌ (Not Applicable)
DimReseller	❌ (Not Applicable)	✅ (Direct Joint)	❌ (Not Applicable)
DimPromotion	✅ (Direct Joint)	✅ (Direct Joint)	❌ (Not Applicable)
DimCurrency	✅ (Direct Joint)	✅ (Direct Joint)	❌ (Not Applicable)

8. Semantic Model Standards & Relationships

To ensure the integrity of calculations within the semantic layer, the following design parameters are strictly enforced:

8.1 Relationship Cardinality

- All relationships between dimension tables and fact tables are established as **One-to-Many (** $1 : \infty$ **)**, where the "One" side is strictly anchored on the primary key of the dimension and the "Many" side resides in the foreign key of the fact table.

8.2 Cross-Filtering Directionality

- Filters MUST flow in a **Single Direction** (from the Dimension to the Fact Table).
- **Bidirectional filtering is prohibited** except under rare, verified circumstances. This prevents ambiguous query paths, unexpected calculation filters, and negative performance impacts.

Correct Single-Direction Flow:

```
┌───────────────────────┐       ┌──────────────────────────────┐
│ DimProduct            ──(1:*)──>  FactInternetSales
│ Filter Source                       Aggregation
└───────────────────────┘       └──────────────────────────────┘
```

8.3 Storage Engine Optimization

- **Storage Mode:** VertiPaq Import Storage Mode for instantaneous in-memory execution.
- **Data Types:** High-precision columns are cast as Decimal or Integer. Avoid storing

high-cardinality alphanumeric fields directly in fact tables to minimize memory overhead.

9. Analytical Scenarios Supported

The structure of this model is optimized to provide high-performance solutions for the operational questions outlined in the BRD:

- **Revenue Analysis (BQ-01 to BQ-03):** Achieved by combining SalesAmount across both FactInternetSales and FactResellerSales, grouped along any attribute hierarchy in DimDate.
- **Channel Performance (BQ-12 & BQ-13):** Comparing InternetSales.SalesAmount against ResellerSales.SalesAmount using dimensions like DimProduct or DimSalesTerritory to contrast market share and relative margins.
- **Sales Representative Quota Analysis (BQ-14 to BQ-16):** Done by linking FactResellerSales.SalesAmount (by representative) to FactSalesQuota.SalesAmountQuota through the conformed DimEmployee and DimDate dimensions, providing clear target attainment percentages.

10. Future Enhancements & Integration Roadmap

To maintain model stability, the design incorporates provisions for seamless horizontal expansion:

```
┌──────────────────────────────────────────────────────────┐
│ Current Model (Sales Domain)                             │
│ [FactInternetSales]    [FactResellerSales]               │
└──────────────────────────────────────────────────────────┘
          │
          │ Conjoined on [DimProduct] Key
          ▼
┌──────────────────────────────────────────────────────────┐
│ Future Expansion (Inventory Domain V2.0)                 │
│ [FactProductInventory]                                   │
└──────────────────────────────────────────────────────────┘
```

10.1 Inventory Analytics Integration (Version 2.0)

- **Fact Table Integration:** Introduction of FactProductInventory containing daily stock counts and warehouse mappings.
- **Shared Dimensions:** Will immediately leverage the existing conformed DimProduct and DimDate dimensions, allowing users to contrast active sales trends against current stock levels without restructuring the core model.